Inclined Planes — Forces on Tilted Axes

SYNOPSIS

A cart on a level rail stays put. Raise one end and it runs away down the ramp. Nothing about that is surprising — we have all ridden a bike up a hill and enjoyed the trip back down, and we all know gravity is responsible. What is worth asking is *how much* of gravity gets through. An incline is free fall with the gravity turned down, and the dial is the angle. That is exactly why Galileo used ramps: they slow gravity to a speed a person with a water clock can actually measure.

BEFORE YOU READ ON — PREDICT

A large adult and a small child each sit on a sled at the top of the same snowy hill, and both are released at the same instant.

Do they slide down at the **same rate**, or does one accelerate harder? Do they reach the bottom together, or does one arrive first?

Write down your answer and one sentence of reasoning before you read on. Galileo was the first person to settle this by measurement rather than argument.

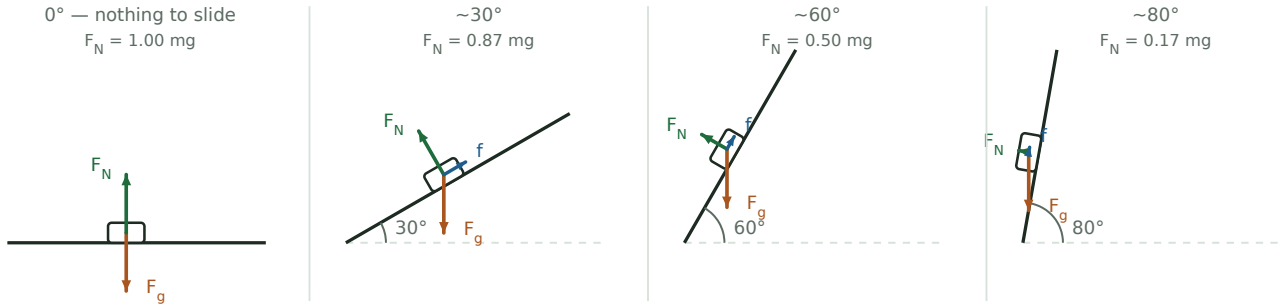
PERFORMANCE EXPECTATIONS

1. **Draw** the free-body diagram for an object on an incline, with the axes tilted to the slope, and say why the normal force needs no resolving while the weight does.
2. **Show** why the angle inside the force triangle equals the incline angle.
3. **Resolve** the weight into components along and perpendicular to the surface, and derive $F_N = mg \cos \theta$.
4. **Derive** $a = g \sin \theta$ for a frictionless incline, and explain why the mass does not appear in it.
5. **Check** the 0° and 90° limits against the level-surface and free-fall results.
6. **Determine** the hanging mass that holds an object in equilibrium on a frictionless incline.

7. **Derive and use** $a = g(\sin\theta - \mu\cos\theta)$ when friction acts, and interpret a negative result.

MOVEMENT ONE · THE FRICTIONLESS INCLINE

Watch the normal force shrink

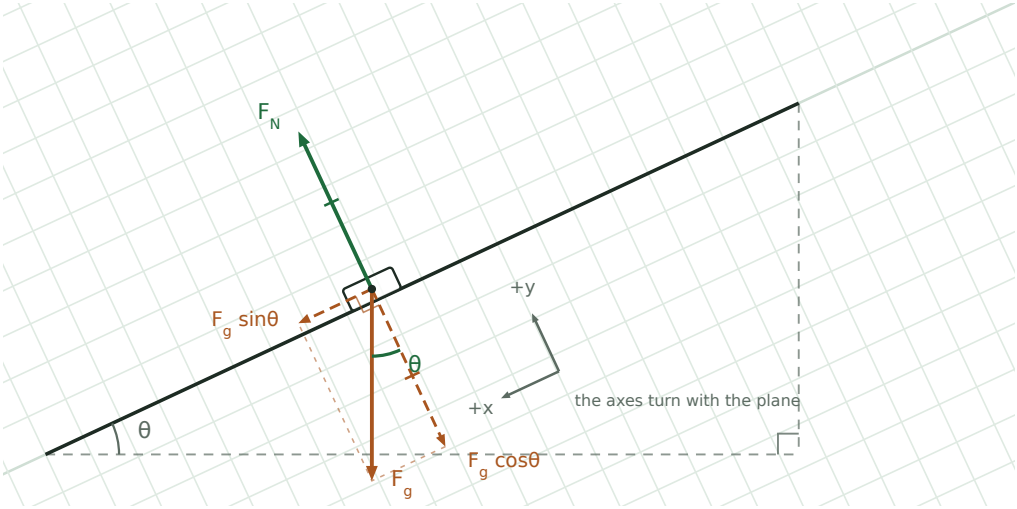


1. At **90 degrees** you are back to two forces. Why? Draw it. At that angle the surface is irrelevant to the motion — how would you describe what the object is doing?

Tilt the axes, not the physics

THE RULE THE WHOLE NOTE RESTS ON

$$\Sigma F_x = ma_x \quad \text{and} \quad \Sigma F_y = ma_y$$



2. Which angle in the force triangle equals θ? Mark it, and **show why** it is the same as the incline angle.

3. Write the component of F_g perpendicular to the surface, and the component along it, both in terms of F_g and θ.

4. There is no net force along y. **Why?** Use that to write F_N in terms of F_g and θ.

The key relationships

Eq. 1 $F_N = mg\cos\theta$

the object stays on the surface, so $\Sigma F_y = 0$ — **true at any angle, with or without friction**

Eq. 2 $F_{g,x} = mg\sin\theta$

the down-slope pull — the only part of the weight that drives the motion

Eq. 3 $a = g\sin\theta$

frictionless only. Note what is missing: the mass

5. The object slides down a frictionless ramp. **Why does it accelerate at all?**

6. Use Newton's second law to show that a = g sin θ. Is this larger or smaller than the free-fall acceleration?

7. Use that equation to explain why the object **is** in free fall when the incline is vertical. How does that square with your physical intuition?

Back to the sled

Holding it still — the measurement

Now connect the object over a smooth pulley to a hanging mass, chosen so the object does not move.

8. Which component of F_g must the hanging weight cancel for the object to be in equilibrium? Explain why it is that one and not the whole weight.

9. The object has a mass of 2 kg and the incline is at 30°. **What hanging mass holds it still?** Work it out first — then set it up and confirm your answer by measurement.

MOVEMENT TWO · NOW PUT FRICTION BACK

A real ramp

Eq. 1 $a = g(\sin\theta - \mu\cos\theta)$

sliding down a real incline — still no mass in it

WORKED — RUBBER ON ICE, $M_K = 0.1$

$\theta = 30°$	$a = 9.8\,(0.500 - 0.1 \times 0.866)$	$= +4.05\ m/s^2$
$\theta = 5°$	$a = 9.8\,(0.087 - 0.1 \times 0.996)$	**$= -0.12\ m/s^2$**

10. Set the ramp at a shallow angle with the block on it, and slowly raise one end until the block **just** breaks loose. Record that angle. What is μ_s for these two surfaces? Compare it with the value you measured by dragging in Note 5 — same block, same surfaces, two completely different methods.

BACK TO YOUR PREDICTION

You predicted whether the adult and the child reach the bottom together. Was it right? Point to the place in $a = g \sin\theta$ that settles it.

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